

SMARTSYMPO 2026

Next-Gen Real-Time Multi-Venue Event Management & Student Routing Platform

System Documentation & Technical Demo Guide

An in-depth specification of platform architecture, database schemas, real-time protocols, conflict-free scheduling algorithms, QR cryptographic gate entry, and campus wayfinding.

PROJECT METADATA

Application Stack:	React 18 (Vite SPA) + Tailwind CSS + HTML5-QRCode + Leaflet.js
Database & Auth:	Supabase Cloud (PostgreSQL 15, Row-Level Security, Realtime CDC)
Microservices:	Node.js / Express Microservice + Nodemailer Gmail SMTP + Web Relays
Target Deployment:	Progressive Web App (PWA) / Mobile-First Campus Deployments
Document Version:	v2.4.0 (Enterprise Academic Release)

EXECUTIVE HIGHLIGHT

SmartSympo eliminates physical queue congestion and double-bookings using atomic PostgreSQL conflict detection, encrypted QR passes with anti-passback defense, live delay broadcasts, and GPS + QR campus wayfinding.

1. Executive Summary & Problem Analysis

Why traditional symposium systems fail and how SmartSympo resolves them

College academic symposiums involve hundreds to thousands of student delegates moving between multiple auditoriums, technical presentation halls, and laboratory tracks simultaneously. Standard pen-and-paper or disjointed Google Form setups exhibit critical operational points of failure:

- Double-Bookings & Schedule Collisions: Delegates register for competing tracks occurring at the exact same hour across distant blocks, leading to unpredicted drop-offs.
- Door Check-in Queues & Proxy Scans: Paper roll calls cause 20-30 minute gate bottlenecks. Static ticket screenshots enable unauthorized pass-sharing.
- Notification Latency: Venue relocations or delayed speaker sessions take too long to propagate, causing empty halls while students wait outside.
- Campus Disorientation: External delegates waste time finding specific laboratories or auditoriums across expansive multi-acre college grounds.
- Coordinator Blindspots: Department faculty have no real-time telemetry on hall fill-rates or missing registered participants.

SmartSympo Core Value Proposition

SmartSympo integrates all symposium operations into an instantaneous, reactive web environment. Built with mobile-first ergonomics, it provides automated conflict-free registrations, dynamic QR door scanning with duplicate detection, real-time stage delay propagation via WebSockets, and "You Are Here" QR landmark campus navigation.

Zero-Config Resilient Dual-Mode Architecture

SmartSympo functions seamlessly connected to live Supabase Cloud PostgreSQL, but also incorporates an automatic, fully persistent LocalStorage and synthetic UUID fallback mode. If campus Wi-Fi falters or cloud credentials are intentionally withheld, all user registration, QR scanning, and hall state management continue operating with zero downtime.

2. System Architecture & Component Design

High-level multi-tier architecture and reactive communication conduits

SmartSympo adopts a layered decoupled architecture where the client layer directly subscribes to PostgreSQL Change Data Capture (CDC) events via WebSockets, minimizing latency while offloading heavy transactions to atomic database RPC functions.

Architectural Layer	Technology	Key Responsibility & Features
Presentation Tier (PWA)	React 18, Vite, Tailwind CSS	Mobile-first PWA, responsive role dashboards, camera QR scanner, audio synthesizers.
State & Context Engine	AppContext.jsx, WebSockets	Central state provider, live alert subscriptions, mock/cloud sync, theme enforcement.
Database & Auth Tier	Supabase (PostgreSQL 15)	Row-Level Security (RLS), GoTrue Auth, Realtime publication channels, atomic stored procedures.
Relay Microservice	Express.js, Nodemailer SMTP	Automated first-login emails, registration PDF/HTML dispatch, WhatsApp & SMS web hooks.
Geospatial Engine	Leaflet.js & OpenStreetMap	Campus vector polygons, department color codes, route polylines, QR landmark parsing.

Real-Time Data Flow (Supabase CDC WebSockets)

When a coordinator triggers a schedule change (e.g. "+10 Minute Delay") or an administrator publishes a Critical Emergency alert, the update is committed to PostgreSQL. Supabase's Realtime engine immediately pushes the diff across active WebSocket channels. All active student and coordinator clients update their interface state in under 120ms without manual page reloads.

Hardware & Native PWA Integration

- **HTML5-QRCode Engine:** High-speed video frame processing with front/rear camera toggling and torch support.
- **Web Audio API Synthesizers:** Custom frequency oscillators producing distinct chimes for valid entry (880Hz), duplicate scans (350Hz buzz), and unauthorized tickets (200Hz).
- **Vibration API Haptics:** Patterned motor vibrations providing physical tactile confirmation to door coordinators in noisy auditorium environments.

3. Database Schema & Relational Models

Production PostgreSQL DDL specifications from supabase/migrations/01_schema.sql

The database enforces relational integrity, duplicate prevention, and non-overlapping registration constraints directly at the database engine level.

Core Relational Tables

Table Name	Primary Keys & Indexes	Description & Integrity Constraints
public.profiles	id (UUID -> auth.users)	Stores full name, role (student, coordinator, admin), roll number, college name, and first_login flag.
public.events	id (UUID)	Symposium events with title, category, hall_number, start_time, end_time, max_capacity, status, delay_minutes.
public.registrations	id (UUID), UNIQUE(student_id, event_id)	Delegate event bookings. Compound unique index prevents double booking of the same event.
public.attendance_logs	id (UUID), INDEX(event_id, student_id)	Immutable audit trail of door scans with check-in timestamp and coordinator UUID.

Clash-Free Atomic Function: register_for_event()

Double-booking protection is enforced atomically via a PL/pgSQL function using SQL OVERLAPS operators:

```
CREATE OR REPLACE FUNCTION
register_for_event(p_student_id UUID,
p_event_id UUID)
RETURNS JSONB LANGUAGE plpgsql AS $$
DECLARE v_start TIMESTAMPTZ; v_end
TIMESTAMPTZ; v_conflict_id UUID;
BEGIN
    SELECT start_time, end_time INTO v_start,
v_end FROM public.events WHERE id = p_event_id;
    -- Atomic Collision Detection across
registered sessions
    SELECT e.id INTO v_conflict_id FROM
public.registrations r
JOIN public.events e ON e.id = r.event_id
WHERE r.student_id = p_student_id AND
(v_start, v_end) OVERLAPS (e.start_time,
e.end_time);
    IF v_conflict_id IS NOT NULL THEN
        RETURN jsonb_build_object('success',
false, 'error', 'TIME_CLASH_DETECTED');
    END IF;
    INSERT INTO public.registrations(student_id,
event_id) VALUES (p_student_id, p_event_id);
    RETURN jsonb_build_object('success', true);
END; $$;
```

4. Role Dashboards & Feature Capabilities

Comprehensive breakdown of Student, Coordinator, and Admin workflows

A. Student Dashboard (/student)

- **Dynamic Chronological Agenda:** Categorizes events into technical paper presentations, hackathons, and guest keynotes with real-time status chips.
- **One-Click Conflict-Free Registration:** Immediate validation against user schedule with clear warning modals if time conflicts exist.
- **Encrypted Dynamic QR Pass:** High-density QR containing encrypted payload { student_id, event_id, pass_code, exp: 24h } with anti-screenshot badge counters.
- **Session Feedback & Rating:** Post-session rating system collecting attendee reviews and peer evaluations.

B. Coordinator Console (/coordinator & /scanner)

- **Hall Venue Selector:** Multi-venue switchboard allowing coordinators to isolate their specific presentation hall.
- **1-Touch Stage Controls:** Immediate stage lifecycle buttons: START EVENT, DELAY 10 MINS, and END EVENT, propagating instantaneously across all student devices.
- **Embedded Door Camera Scanner:** Instant badge detection with anti-passback defense (amber warning and buzz for duplicate entries).
- **Missing Attendees Drawer:** Live roster showing registered students who haven't checked in yet, complete with 1-tap nudge broadcast triggers.

C. Admin Analytics & Operations Console (/admin)

- **Real-Time Venue Occupancy Heatmap:** Live visual cards monitoring hall capacity vs check-in counts with warning thresholds.
 - **Event Creation & Hall Mapping:** Intuitive form to allocate new symposium events to physical halls and set capacity limits.
 - **System-Wide Emergency Broadcast:** Multi-venue critical alert trigger with red siren banner and admin-only override authority.
 - **Roster & Analytics Export:** 1-click export of verified attendance records to Excel (.xlsx) and CSV formats.
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5. Campus Wayfinding & Landmark Scanner

Spatial navigation engine for multi-acre university campuses

SmartSympo incorporates a custom vector-geonavigation layer designed to eliminate the common issue of lost attendees wandering across sprawling campus facilities.

Key Navigation Components (CampusMap.jsx & LandmarkScannerModal.jsx)

1. Interactive Campus Maps: Renders real GPS boundaries, building polygons, parking lots, department blocks, and seminar halls using Leaflet & OpenStreetMap.
2. "You Are Here" Physical QR Plaque Scanning: Physical QR stickers placed on corridor pillars and main gates. Attendees scan the plaque using their phone camera to instantly localize their position on the map.
3. Dynamic Directional Polyline: Upon detecting a landmark, the system computes the vector route and renders an illuminated polyline leading directly to the user's destination hall.
4. Printable Landmark Badge Generator: Organizers can generate and print formatted QR badges for campus pillars with one click from within the application interface.

6. AI Chatbot & Communication Microservice

Context-aware intelligence and multi-channel notification dispatch

SmartSympo Virtual Concierge (Chatbot.jsx)

A floating AI assistant with pre-programmed quick action chips ("Where is Hall A?", "Today's Event Schedule", "Live Alerts", "QR Check-in Guide"). It parses symposium events, venues, and live delays to provide instant conversational answers to attendees 24/7.

Notification Microservice (server/index.js)

A dedicated Node.js Express server running on port 5000 offering automated background dispatch services:

- /api/send-welcome-email: Delivers professional HTML welcome emails upon student registration via Gmail SMTP.
 - /api/send-event-confirmation: Dispatches confirmed event passes with scheduling details and venue directions.
 - /api/relay-whatsapp & /api/relay-sms: Web hooks enabling direct message forwarding to WhatsApp and SMS gateways for emergency notifications.
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7. Comprehensive Technical Demo Script

Step-by-step presentation guide for academic evaluators, reviewers, and live audiences

This structured 6-step walkthrough demonstrates every technical feature and failure-handling capability of SmartSympo during a live presentation:

Step 1: Role Authentication & Switchboard

Open `http://localhost:5173`. Display the login screen. Point out the quick seed credentials for Admin (admin / 2005), Coordinator (coordinator / 2005), and Student (student / student123). Log in as Student.

Step 2: Clash-Free Registration & QR Pass Generation

Navigate to the event catalog. Register for "Keynote Address: Future of Generative AI". Notice the immediate confirmation. Attempt to register for a second event occurring during the exact same time slot to demonstrate the collision prevention rejection. Open "View QR Pass" to reveal the dynamic encrypted badge.

Step 3: Door Scanner & Anti-Passback Defense

Switch to Coordinator view (`/coordinator`) and filter by "Hall 1 (Main Auditorium)". Launch the QR Camera Scanner (`/scanner`). Point the camera at the student QR pass (or select it from the Dev Simulator). Observe the green checkmark and 880Hz audio chime. Scan the same ticket a second time: demonstrate the duplicate scan warning and error buzz.

Step 4: Real-Time Hall Controls & Delay Broadcast

As Coordinator, click "+10 Minute Delay". Switch to the Student view: point out how the Live Alert Banner updates instantaneously across all devices without reloading the browser, driven by PostgreSQL CDC WebSockets.

Step 5: Campus Geonavigation & Landmark Scanner

Open `/navigation`. Show the interactive campus building polygons. Open the Landmark Scanner Modal, select a campus pillar from the simulation list (or scan a physical badge), and watch the map orient to "You Are Here" and render a directional polyline to the venue.

Step 6: Admin Occupancy Heatmap & Emergency Siren

Log in as Admin (`/admin`). Inspect the real-time capacity fill gauges. Open Emergency Broadcast Modal, type a high-priority alert, and broadcast it. Observe the red emergency siren banner across all interfaces, then show the admin-only clear authority.

8. Deployment, Configuration & Run Guide

Local execution steps and environment variable configuration

Environment Configuration (.env)

Ensure the following environment variables are set in client/.env and server/.env:

```
# Client Environment (client/.env)
VITE_SUPABASE_URL=https://your-project-id.supabase.co
VITE_SUPABASE_ANON_KEY=your-supabase-anon-key

# Server Environment (server/.env)
PORT=5000
GMAIL_USER=your-symposium-email@gmail.com
GMAIL_APP_PASSWORD=your-gmail-16-char-app-password
```

Execution Commands

Command	Working Directory	Description & Expected Behavior
npm run dev	smart-sympo/	Launches root concurrent dev runner for client and backend.
npm run dev:client	smart-sympo/client	Starts Vite dev server at http://localhost:5173 with HMR.
npm run server	smart-sympo/server	Launches Express notification microservice on port 5000.
npm run build	smart-sympo/client	Executes production Vite TypeScript/JSX bundling.

CONCLUSION & OPERATIONAL READINESS

SmartSympo represents a fully tested, production-grade symposium ecosystem. It combines zero-latency database synchronization, high-speed camera scanning, anti-tamper security, and geospatial navigation into a frictionless, elegant user experience.

